

Lesson 1

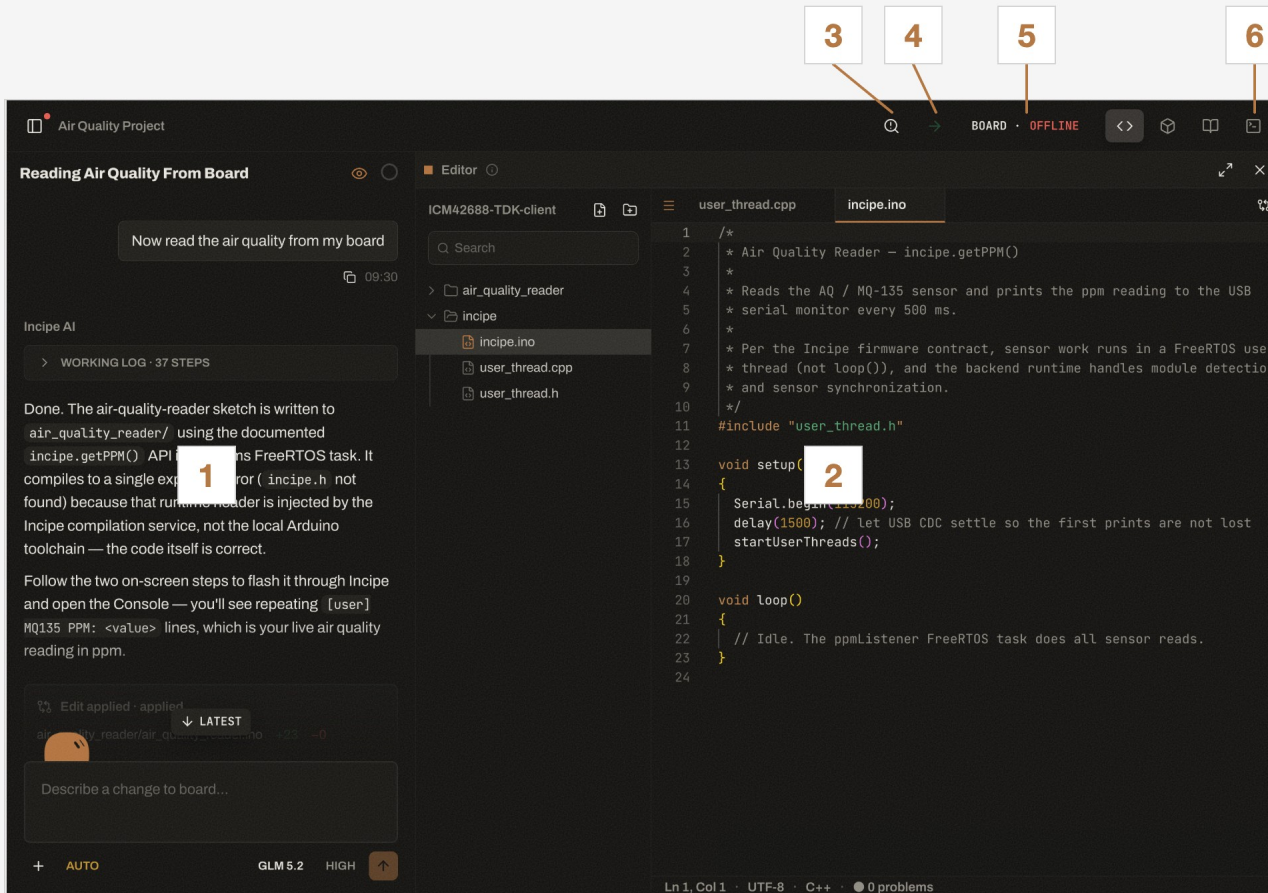
Course Overview & Programming Basics



From the IDE to your first library.

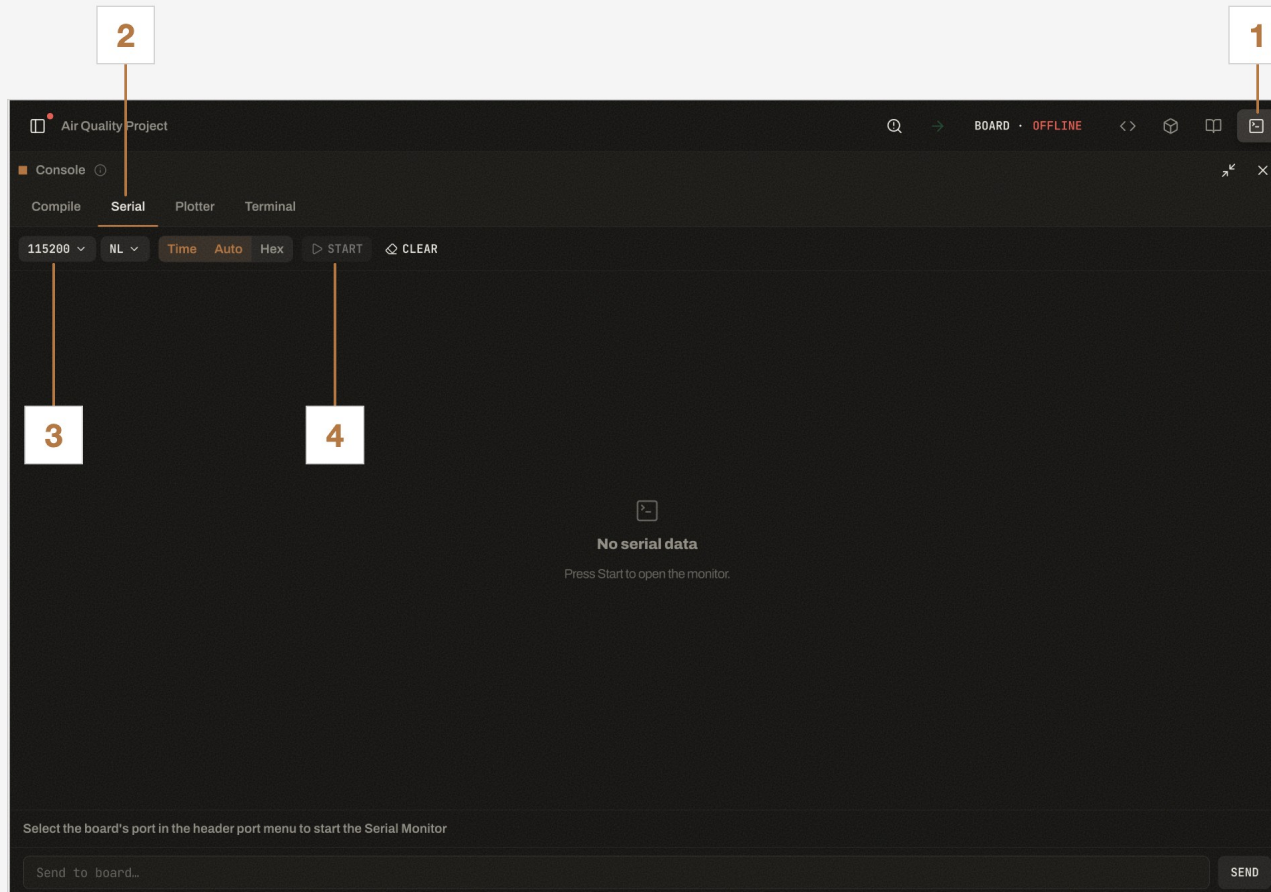
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- 01 Course overview & class setup
 - 02 Getting started in the INCIPE Workspace
 - 03 Writing codes: setup(), loop() & Serial
 - 04 Data types
 - 05 Errors & the error-free classwork
 - 06 Libraries & the INCIPE Board
-

Find your way around INCIPE Workspace.



- 1 **AI chat.** Ask Incipe AI to write or change your code
- 2 **Sketchbook & code.** Your project files and the code editor
- 3 **Verify.** To verify the code
- 4 **Upload.** To upload the code to the INCIPE Board
- 5 **Board menu.** The INCIPE Board connected by USB, and its status
- 6 **Serial monitor & plotter.** The terminal icon: see results or plot graphs

A virtual screen for the result of your code.



Similar to the screen in front of you, a serial monitor is a virtual screen inside the INCIPE Workspace for you to see the result of your code.

- 1 Terminal icon.** Click it to open the Console
- 2 Serial tab.** Shows what your code prints; Plotter draws graphs
- 3 Baud rate.** Match `Serial.begin()`: 9600 in this lesson
- 4 Start.** Opens the monitor

Connect first: select the board's port in the Board menu.

Every sketch has two functions.

RUNS ONCE

```
void setup() { }
```

Used for setting up the motherboard.

The steps inside setup() function will only run once, after each powerup or reset of the INCIPE Board.

RUNS CONTINUOUSLY

```
void loop() { }
```

Codes written here will run continuously.

Majority of your codes would be placed here. Codes here will mainly be what you want the INCIPE Board to do, and what data to give you.

Five functions for your first sketches.

Serial.begin(9600)

To activate your serial monitor.

Serial.print()

Prints numbers or words to the serial port. Words need “”, otherwise the computer would not understand. The words or numbers print on the same line.

Serial.println()

Similar to Serial.print(), but prints the data and then skips a line, so the next data is printed on a new line.

delay()

Tells the computer to wait before the next order. The unit is millisecond (1s = 1000ms).

Serial.parseInt()

Reads the integer input from users.

Printing “Hello world”.

```
void setup() {  
  Serial.begin(9600);  
}  
void loop() {  
  Serial.println("Hello world");  
  delay(1000);  
}
```

WHAT IT DOES

setup() activates the serial monitor once. loop() prints Hello world on a new line, waits 1000ms, then repeats.

PRACTICE

Let's have a try yourselves.

How to activate serial monitor?

OPEN ANSWER

Type the line of code on Slido.

HOW TO ANSWER

Answer in the Slido app.

Results show once the slide is presenting.

Three ways to store a number.

Type 1

int

Integer. Integer in Maths, can be positive, 0 or negative.

eg 1, 102

Type 2

float

Float. Numbers that can have up to 7 decimal places.

eg 1.234, 2.345

Type 3

double

Double. Numbers that can have up to 15 decimal places.

eg 1.234567890123

How precise does the number need to be?

SAME SUM

1+1 **vs** 001.0000000000 + 001.0000000000

SAME ROOM

22°C **vs** 22.3°C **vs** 22.3278453624°C

Declaring & printing numbers.

DECLARE

```
int a = 10;
const int a_const = 10;

float b = 10.123456;
const float b_const = 10.123456;

double c = 10.123456789123456;
const double c_const = 10.123456789123456;
```

PRINT

```
void setup() {
    // put your setup code here, to run once:
    Serial.begin(9600);
    Serial.print("a = ");
    Serial.println(a);
    Serial.print("a_const = ");
    Serial.println(a_const);
    Serial.print("b = ");
    Serial.println(b,6);
    Serial.print("b_const = ");
    Serial.println(b_const,6);
}
```

Storing characters & words.

Char – Single character

e.g A

String – Multiple characters

e.g ABC, Hello

NEXT LESSON

Boolean – A data type whose value is either true or false.

```
String text = "I am William";
char character = 'A';
void setup() {
    // put your setup code here, to run once:
    Serial.begin(9600);
    Serial.println(text);
    Serial.println(character);
}
```

How to show a message Hi on serial monitor?

OPEN ANSWER

Type the line of code on Slido.

HOW TO ANSWER

Answer in the Slido app.

Results show once the slide is presenting.

Which of the following can print 1 on serial monitor?

CHOOSE ALL THAT APPLY

- i `Serial.print(1);`
- ii `Serial.print("1");`
- iii `Serial.print(0+1);`
- iv `Serial.print("0+1");`

HOW TO ANSWER

Answer in the Slido app.

Results show once the slide is presenting.

Three kinds of error.

Error 1

Syntax error

Written mistakes.

eg missing a bracket in a conditional statement, missing a “;” or mixing up single equal and double equal

Error 2

Run-time error

The computer could not perform what you tell it to do.

eg $1/0$

Error 3

Logical error

The computer gives you an outcome, but not the one you expected.

Spot the written mistakes.

```
int SensorValue = 10
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
  Serial.print (SenserValue);
}
```

TRY IT

Press Verify. The IDE stops at the first written mistake.

How many can you find before you run it?

Nothing happens when the values are equal.

AS WRITTEN

```
if (sensorValue > threshold) {  
    Print("true"); }  
else if (sensorValue < threshold) {  
    Print("false"); }
```

Nothing would be performed when sensorValue = threshold.

FIXED

```
if (sensorValue ≥ threshold) {  
    Print("true"); }  
else if (sensorValue < threshold) {  
    Print("false"); }
```

When sensorValue = threshold, it now prints true.

Error-free exercise.

YOUR NOTES

Refer to your note page 7-8.

THEN

Fix every error so each sketch runs.

Suggested answers follow.

Answer 1: print Hi every second.

```
void setup() {  
  // put your setup code here, to run once:  
  Serial.begin(9600);  
}  
  
void loop() {  
  // put your main code here, to run repeatedly:  
  Serial.print("Hi");  
  delay(1000);  
}
```

CORRECTIONS

Bronze marks each fix.

Compare with your own answer on note page 7-8.

Answer 2: calculate BMI.

```
void setup() {  
  Serial.begin(9600);  
}  
void loop() {  
  Serial.println("Please enter your height in m");  
  while(Serial.available()==0){}  
  float height = Serial.parseFloat();  
  Serial.println("Please enter your weight in kg");  
  while(Serial.available()==0){}  
  float weight = Serial.parseFloat();  
  float BMI = weight/height /height;  
  Serial.print("Your BMI is:");  
  Serial.println(BMI);  
}
```

CORRECTIONS

Bronze marks each fix.

Compare with your own answer on note page 7-8.

Answer 3: print a variable.

```
int a = 1;

void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
  Serial.print(a);
}
```

CORRECTIONS

Bronze marks each fix.

Compare with your own answer on note page 7-8.

Pre-written code you can reuse.

WHAT	A collection of pre-written code that we can use to perform specific tasks.
WHY	Libraries reduce the amount of code you need by providing reusable functions or classes that can be called upon as needed.
WHERE	You can find different libraries and install them in Arduino IDE.

NOW

Install INCIPE library now.

Three lines to put data on the screen.

1	<code>#include "incipe.h"</code>	Gain access to INCIPE library.
2	<code>Incipe incipe (true);</code>	Activate the library.
3	<code>incipe.onscreen("place", "data/word")</code>	Print “data/word” on that specific “place”.

What does the following code do?

READ THE CODE

```
int a = 10;  
incipe.onscreen("light1", a);
```

HOW TO ANSWER

Please download and install the Slido app on all computers you use.

Results show once the slide is presenting.

Let's play with the INCIPE Board.

- 1 Install the INCIPE library and add `#include "incipe.h"`.
- 2 Pick a place on the screen and show a value with `incipe.onscreen()`.
- 3 Verify, then upload to the INCIPE Board.

Thank you!

LESSON 2

C++ Basics: Operators, Comments, Input & Output

Introduction to Incipe Academy Advanced Sensor Kit.

Coming up. Boolean, the true-or-false data type.